
Corrigé — Exercice 15 • Bac principal 2026

1) $\det(A) = -3(9 - 4) - 2(-6 - 4) + 2(4 + 6) = -15 + 20 + 20 = 25 \neq 0 : A$ inversible.

$$A^2 = \begin{pmatrix} 17 & -8 & -8 \\ -8 & 17 & -8 \\ -8 & -8 & 17 \end{pmatrix}, \text{ et } A^2 + 4A = 5I_3. \text{ D'où } A(A + 4I_3) = 5I_3, \text{ donc } A^{-1} = \frac{1}{5}(A + 4I_3) =$$
$$\frac{1}{5} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}.$$

2) (S) s'écrit $AX = \begin{pmatrix} -10 \\ 10 \\ 20 \end{pmatrix}$, d'où $X = \frac{1}{5} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} -10 \\ 10 \\ 20 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 50 \\ 30 \\ 20 \end{pmatrix} = \begin{pmatrix} 10 \\ 6 \\ 4 \end{pmatrix}.$

$$(x, y, z) = (10, 6, 4)$$